

Answer Key — Biology Grade 11

Sub-Strand 1.3: Taxonomy II • Lesson 1: Why classify organisms?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	B	A	A	D	B	C

USE AFTER: PREDICT PHASE — Predict: learners write a prediction on how many groups, what rule, and whether their groups would match their neighbour's

Q1. Four learners will each group the same 13 organisms (maize, bean plant, grass, fern, moss, mushroom, mosquito, butterfly, earthworm, snail, chicken, lizard and goat). Amina uses size, Brian uses colour, Cheche uses where the organism lives, and Dennis uses whether people eat it. What is the most reasonable prediction about their groupings?

- A. The groupings will match as long as the learners work in the same classroom
- B. The groupings will probably differ because each learner is using a different rule**
- C. All four groupings will be exactly the same because the organisms are the same
- D. Only Amina's grouping can work because size is always easy to measure

Correct answer: B. Each learner uses a different personal rule (size, colour, habitat, food value), so different groupings are expected even though the organisms are the same.

USE AFTER: OBSERVE PHASE — Observe: learners record at least two observable features for each of the 13 organisms

Q2. During the observation of the 13 organisms, learners must record features they can see for themselves. Which of these is an observable feature of the butterfly?

- A. It is less important than the goat
- B. It has wings**
- C. It is harmful to people
- D. It is useful as food

Correct answer: B. Having wings can be seen directly by looking at the butterfly. Harmful, useful as food and importance are opinions or uses, not observed body features.

USE AFTER: OBSERVE PHASE — Observe: teacher asks 'Why did you put the mushroom with the plants? What made you decide that?'

Q3. While grouping the organisms, a learner says: 'The mushroom is a plant because it grows in soil.' At this point in the lesson, what is the best response from the teacher?

- A. Point out that 'grows in soil' is a habitat-based rule, and ask what observable features of the mushroom could be compared with the other organisms**
- B. Say that the mushroom is an animal because it cannot move
- C. Agree, because anything that grows in soil is a plant
- D. Tell the learner to stop asking about the mushroom

Correct answer: A. Where an organism lives is only one everyday rule. The teacher should push learners to give observable features as evidence, and the mushroom's group is left as a question to explore in later lessons.

USE AFTER: EXPLAIN PHASE — Explain: gallery walk comparing groupings by colour, size, habitat or usefulness

Q4. One group places the goat and the chicken together under the title 'farm animals'. Another group places them in different groups because the goat has hair and the chicken has feathers. Both groups were looking at the same two organisms. Why did their groupings differ?

- A. The two groups used different rules to make their groups**
- B. One group made a mistake in observing and the other group observed correctly
- C. Groupings differ only when learners come from different counties
- D. The goat and the chicken are not both living things

Correct answer: A. The first group used usefulness or where the animals are kept, while the second used body covering. The different results come from the different rules, not from wrong observation.

USE AFTER: EXPLAIN PHASE — Explain: Think-Pair-Share 'Why do scientists need a standard system of classification?'

Q5. After the gallery walk, a learner from Kericho says: 'I can use your group's manila paper to place the mushroom exactly where you did.' Your group grouped by 'what looks useful to us'. Why might the learner from Kericho not get the same result?

- A. Because groupings only work in the school where they were made
- B. Because people in Kericho cannot see the same organisms
- C. Because the mushroom cannot be grouped at all
- D. Because a rule such as 'useful to us' can mean different things to different people, so the groupings will not match**

Correct answer: D. Rules such as usefulness, colour or size are judged differently by different people. This shows why a standard system based on shared, observable characteristics is needed, so anyone anywhere gets the same result.

USE AFTER: DQB PHASE — DQB: learners post questions under the driving question 'How can we identify and classify living organisms accurately?'

Q6. The class is adding questions to the Driving Question Board under 'How can we identify and classify living organisms accurately?' Which question fits best on this board?

- A. Which organism in the display is the cheapest to buy?
- B. How do scientists decide which features to use when they classify organisms?**
- C. Which organism in the display is the biggest?
- D. Which group made the neatest manila paper?

Correct answer: B. This question is directly about the process of classification and about choosing characteristics. It is the kind of question that leads to the next lesson. The other questions are about size, price or neatness.

USE AFTER: END OF LESSON — Model: exit ticket (b) give one observable characteristic to group the maize, bean plant and grass together

Q7. For the exit ticket, a learner must give one observable characteristic that could be used to group the maize, the bean plant and the grass together. Which answer is the best?

- A. They are all eaten by people
- B. They all grow in Kisumu
- C. They all have leaves and roots**
- D. They are all exactly the same size

Correct answer: C. Having leaves and roots is an observable feature shared by all three. Grass is not normally eaten by people, the three differ in size, and growing in Kisumu is not a body feature.